

A REVIEW ON SMOTE-MRS: A MULTIREOLUTION SAMPLING APPROACH FOR HANDLING IMBALANCED MEDICAL DATASETS IN ANEMIA PREDICTION

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Abstract—The SMOTE-Multiresolution Sampling (SMOTE- MRS) framework represents a recent innovation in address- ing the challenges of class imbalance within medical datasets, specifically in anemia prediction. This review evaluates the methodology, experimental design, comparative performance, and limitations of SMOTE-MRS proposed by Saputra *et al.* (2024). The hybrid model integrates K-Means clustering, Syn- thetic Minority Over-sampling Technique (SMOTE), and Ran- dom Over Sampling (ROS) to generate balanced training data. Through this synthesis, SMOTE-MRS achieves improved diag- nostic accuracy and robustness across multiple datasets including anemia, diabetes, breast cancer, and kidney failure. This paper critically analyzes the algorithm’s contributions, compares it with existing oversampling methods such as SMOTE, SMOTE-ENC, and KNSMOTE, and discusses future improvements for large- scale and deep learning applications.

Index Terms—SMOTE-MRS, class imbalance, anemia predic- tion, K-Means clustering, medical datasets, machine learning.

INTRODUCTION

The prevalence of anemia remains a major global health issue, affecting approximately 1.62 billion people worldwide, or nearly 25% of the global population [1]. In Indonesia, the situation mirrors global trends, with rates of anemia in preg- nant women and children exceeding 30% in recent years [2]. Machine learning models offer new tools for detecting anemia; however, their performance is hindered by highly imbalanced datasets, where the majority (healthy) class overshadows the minority (anemic) class.

Traditional learning algorithms tend to overfit the majority class, leading to misclassification of the minority class, which is critical in clinical decision-making [3]. Common sampling methods such as SMOTE [4], SMOTE-ENC [5], and Random Over Sampling (ROS) [6] attempt to alleviate this problem but face challenges like overfitting, noisy synthetic points, or bias towards dominant features.

The SMOTE-MRS model, introduced by Saputra *et al.* [7], provides an innovative hybrid strategy that enhances minority class representation through multi-resolution sampling. This review examines its architecture, experimental evaluation, and broader implications in healthcare analytics.

Class imbalance remains a key obstacle in medical prediction models, where minority cases (patients with disease) are outnumbered by healthy cases. Various oversampling and hybrid strategies have been proposed to overcome this imbalance. This section reviews existing works related to SMOTE, its variants, and emerging generative models.

A. *SMOTE and Its Variants*

SMOTE [4] is a foundational approach for generating synthetic minority samples by interpolating between minority class neighbors. Variants such as Borderline-SMOTE, SMOTE-Tomek, and SMOTE-ENN refine this process by cleaning noisy samples and reducing class overlap. However, these methods remain sensitive to data distribution and often fail when the minority class forms multiple small clusters [10].

B. *Cluster-based Oversampling: SMOTE-MRS*

Cluster-aware oversampling methods divide minority data into subgroups to preserve local structures before applying oversampling. The SMOTE-MRS algorithm [7] extends this principle using K-Means clustering, SMOTE, and Random Oversampling (ROS). This multiresolution strategy generates synthetic samples within homogeneous regions, improving minority class diversity and classifier performance. Experimental results showed notable accuracy and AUC improvements in anemia, diabetes, and cancer datasets.

C. *Generative Approaches: VAE and GAN-based Models*

Recent research explores the use of deep generative models for minority sample synthesis. Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs) can learn the probability distribution of minority data and generate realistic synthetic samples [11], [12]. Although GAN-based approaches provide high-quality samples, they require significant computational resources and careful tuning to avoid instability. Lightweight VAEs provide a balance between realism and efficiency, motivating the hybrid approach explored in later works such as AHO-SMOTE.

D. *Applications in Medical Prediction*

Several studies applied SMOTE variants in medical contexts. Yimer *et al.* [13] utilized SMOTE with Random Forest for anemia detection using EDHS data, achieving improved recall. Other works [10], [14] compared SMOTE-ENN and SMOTE-Tomek on diseases such as diabetes and heart disease, confirming consistent gains in minority-class prediction.

E. *Identified Research Gaps*

From prior works, several gaps remain unaddressed:

- Most SMOTE variants rely on linear interpolation without learning data distributions.
- Cluster-based SMOTE methods like SMOTE-MRS are rarely validated across diverse datasets.
- Generative models are computationally heavy, limiting clinical deployment.
- Few methods evaluate synthetic data realism or interpretability.

F. *Position of the Current Study*

The current review builds upon SMOTE-MRS by analyzing its hybrid structure and limitations, setting the foundation for lightweight generative oversampling such as AHO-SMOTE, which integrates clustering with a Variational Autoencoder for efficient minority synthesis.

OVERVIEW OF THE SMOTE-MRS FRAMEWORK

The SMOTE-MRS technique integrates three mechanisms:

- 1) **K-Means Clustering:** Divides the dataset into k homogeneous clusters to preserve local data structures.

2) SMOTE (Synthetic Minority Oversampling Technique): Generates synthetic minority samples within each cluster using nearest-neighbor interpolation.

3) Random Over Sampling (ROS): Duplicates existing minority samples to further balance the data distribution.

The model was trained and tested using Random Forest (RF), Naïve Bayes (NB), and Support Vector Machine (SVM) classifiers under 10-fold cross-validation. On anemia datasets from Yogyakarta, Indonesia, SMOTE-MRS achieved an accuracy of 0.973, recall of 0.990, precision of 0.968, F1-score of 0.979, and AUC of 0.994, outperforming other methods [7].

COMPARATIVE ANALYSIS

KNSMOTE [8] combined K-Means and SMOTE for decision tree enhancement but lacked random oversampling support. G-SMOTENC [9] targeted mixed-type data (nominal and continuous) using gradient-based resampling. Compared with these, SMOTE-MRS offers multi-resolution clustering that allows more granular synthesis of minority samples.

Empirical evaluations demonstrated that SMOTE-MRS consistently surpassed classical methods (SMOTE, SMOTE-ENC, and ROS) across multiple datasets—diabetes, breast cancer, and chronic kidney disease. Particularly in anemia diagnosis, SMOTE-MRS improved recall without sacrificing precision, crucial for minimizing false negatives in clinical diagnosis.

CRITICAL INSIGHTS AND LIMITATIONS

Despite its high accuracy, SMOTE-MRS faces several technical limitations:

- Dependence on K-Means: Sensitive to cluster initialization and non-linear data distributions.
- Computational Cost: Higher runtime (0.27s vs. 0.20s for SMOTE) due to clustering overhead.
- Parameter Sensitivity: Requires tuning of cluster count and neighborhood size.
- Domain-Specific Constraints: Limited validation on image-based or high-dimensional medical datasets.

Future enhancements may incorporate adaptive clustering, density-based oversampling, and deep learning integrations such as autoencoders or GANs.

FUTURE PROSPECTS

SMOTE-MRS can be integrated into deep neural architectures to enable joint feature learning and resampling. Future developments could include automated hyperparameter optimization, distributed computing adaptation, and applications in real-time healthcare monitoring.

CONCLUSION

The SMOTE-MRS model represents a significant advance in handling imbalanced medical datasets. Its hybrid architecture enhances minority class representation, yielding substantial performance gains in anemia prediction and similar diagnostic challenges. However, further optimization for scalability and deep learning compatibility is recommended.

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